

User Story	L1 – Raspberry Pi	to	Week 47
Informal description	As a technician		
	... I would like to install and configure a Raspberry Pi		
	... so that sensors can be read and their values displayed on the screen and sent to a web server.		
Requirements according to the work order	Must • The Raspberry Pi is mounted on the screen and the screen is connected. • The "Raspberry Pi OS" operating system is installed on the Raspberry Pi. • The screen orientation (landscape) is set correctly. • A static IP address (Ethernet) and a host name (IAM code) are configured. • The I²C interface is enabled. • The supplied files have been copied to the correct folders and the configuration file has been adjusted. • The firmware starts automatically during boot. • Docker is installed. • The supplied web page for displaying the current measurements is hosted by a web server in a Docker container. • The website is displayed in kiosk mode. • <u>Documentation</u>: The relevant installation and configuration steps with all necessary commands and explanations.		
	Should • The Raspberry Pi is accessible via SSH. • The current measurements are displayed in an appealing manner.		
	Could • WLAN is set up and configured correctly. • SSH access is not possible with a password. • The measurement data history is displayed.		
Tasks according to the work order			
• Install "Raspberry Pi OS (64-bit)" on the microSD card. • Configure the operating system as specified.			

- Activate the I²C interface [1].
- Install Docker via script (get.docker.com, [2]). Docker must later function without `sudo` [3].
- Create the folder `/opt/station`
- Copy the supplied files (firmware including configuration file, website, and scripts for testing the sensors) to the Raspberry Pi.
 - Copy all files for the website to the folder `/opt/station/www`.
 - Copy all files for the firmware, configuration file, and scripts for testing to the `/opt/station` folder
- Adjust the configuration file (`/opt/station/config.toml`) according to the work order
(In particular, the serial number and I²C addresses will be configured later in the Atelier User Stories .)
- Create Docker container (php:8.4-apache) [4]:

```
$ docker run -d -name php -p 8080:80 -v /tmp/station:/tmp/station -v opt/station/www:/var/www/html php:8.4-apache
```
- Configure autostart [5].
 - To do this, use the file `~/.config/labwc/autostart`. This is executed as a shell script after startup. `~` refers to the home directory of the logged-in user
 - Create the folder `/tmp/station`.
 - Start the script (firmware) for reading and sending the sensors in the background (use `&`) and redirect the standard output to the file `/tmp/station/result.csv`.
Attention: This is a Python3 script!
 - Start the container (name: php) automatically.
The supplied website is now accessible at <http://localhost:8080>.
 - Automatically start the browser in full-screen mode (kiosk) and display the current measured values.

Resources

- [1] <https://www.raspberrypi.com/documentation/computers/configuration.html#raspi-config>
- [2] <https://docs.docker.com/engine/install/debian/>
- [3] <https://docs.docker.com/engine/install/linux-postinstall/#manage-docker-as-a-non-root-user>
- [4] <https://docs.docker.com/engine/storage/bind-mounts/>
- [5] <https://www.raspberrypi.com/tutorials/how-to-use-a-raspberry-pi-in-kiosk-mode/>